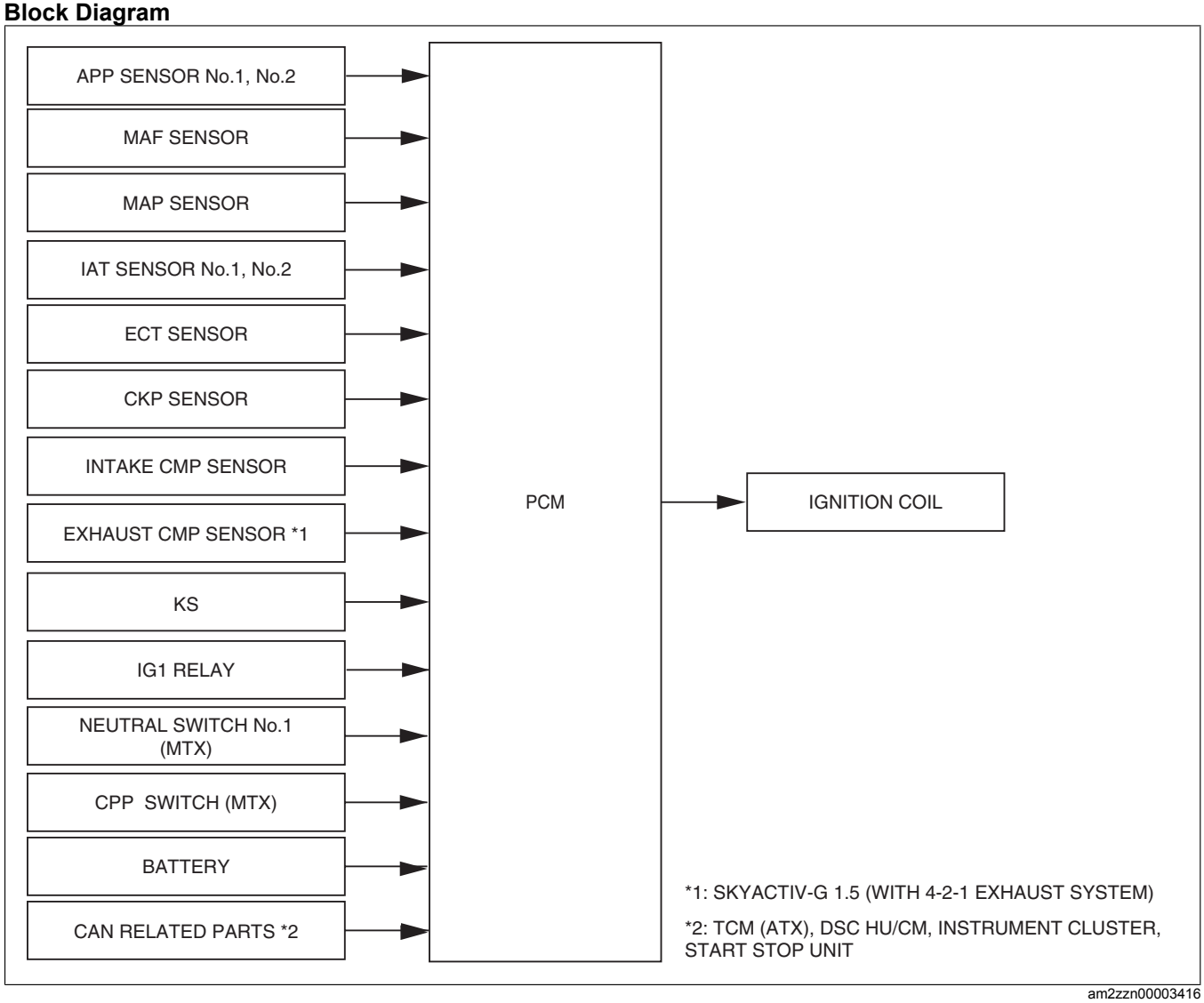
ELECTRONIC SPARK ADVANCE CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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Outline

- Controls ignition for optimum timing (according to engine operation conditions).
- Serviceability has been improved by eliminating the necessity of ignition timing adjustment.
- The PCM determines the engine operation conditions (based on input signals from each sensor) and blocks current to the ignition coils at the calculated ignition timing. This causes the spark plugs to discharge (ignite) by the effect of electromagnetic mutual induction.

Block Diagram



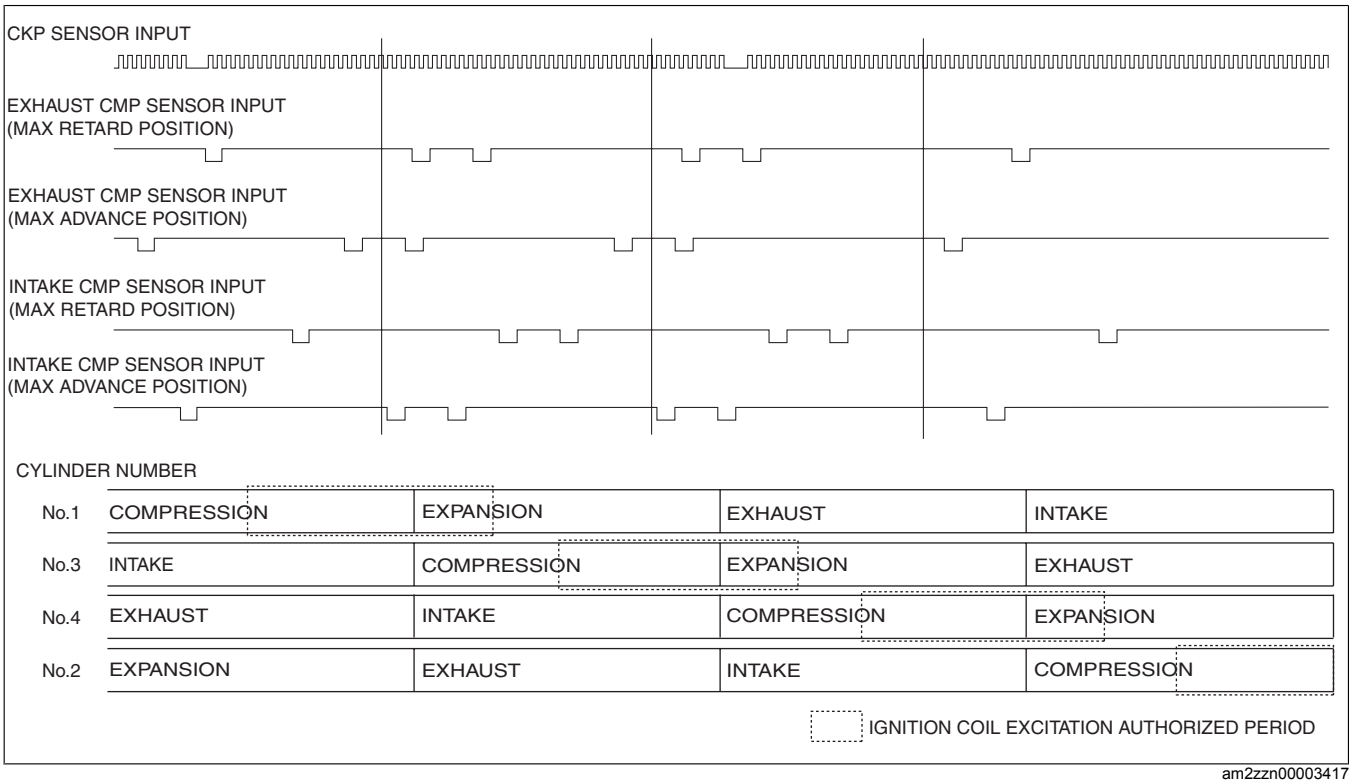
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Operation

Ignition method

- The PCM energizes the ignition coils according to the ignition timing calculated from the engine operation conditions and the igniter energization time.
- The igniter energization time (ignition coil energization time) is determined according to battery voltage and engine speed.

Timing chart



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Determination of ignition timing

Division of control zones

- The PCM performs different ignition controls for engine starting and those other than engine starting to stabilize the engine starting.

Control zone	Control condition	Ignition method
Engine start zone	Engine speed is **less than 500 rpm**	Engine starting
Normal zone	Engine operation except start zone	Cycle estimated ignition (Determines ignition timing adding each correction to basic spark advance)

Ignition timing calculation method table

A: Ignition timing base, B: Correction for ignition timing

Contents		Calculation method or determination method for ignition timing, spark advance, and correction	Control zone	
			ENGINE START ZONE	NORMAL ZONE
Starting ignition		Determination based on engine coolant temperature	A	
Cycle estimated ignition	Basic spark advance	Set value according to engine speed and charging efficiency*		A

Correction	Engine coolant temperature spark advance correction	**According to engine coolant temperature** Purpose: Ensures combustion stability when engine coolant temperature is low • High charging efficiency*, low engine coolant temperature→large correction		B
	Intake air temperature correction	**According to engine intake air temperature and engine coolant temperature** Purpose: Suppresses knocking when engine intake air temperature is high • High engine intake air temperature, high engine intake air temperature →large correction		B
	Warm-up promotion spark retard correction	**Maximum 50 s after engine start** Purpose: Activates the catalytic converter earlier • According to engine coolant temperature→correction		B
	Torque reduction correction	**According to torque reduction request from TCM, or DSC HU/CM** Purpose: Such as reduction of shift shock, traction control, or suppression of vehicle vibration • Large torque reduction request→large correction		B
	Knocking spark retard correction	**Acceleration when charging efficiency* volume increase (acceleration amount) is given value or more** Purpose: Knocking suppression • Large acceleration amount→large correction		B
	Valve timing correction	**When phase difference changes due to electric variable valve timing control (SKYACTIV-G 1.5 (with 4-2-1 exhaust system)) and hydraulic variable valve timing control** Purpose: Ensures combustion stability • Correction according to change in phase difference		B

* : Charging efficiency is the ratio of the actual amount of intake air to the maximum air charging amount (mass volume) of the cylinder. This value increases proportionately to the increase in engine load.

- If immobilizer system related information (engine start prohibited) is received from the start stop unit, the PCM stops ignition.
- When a collision signal is received from the SAS control module, the PCM stops the ignition.